

Solve each system using Gauss Elimination

1) $3x + 2y - 4z = 5$

$$x - 2y + z = 4$$

$$2x + y - 2z = 4$$

2) $x + 2y - z = 3$

$$2x + y - 3z = 2$$

$$x + 3y - 4z = 1$$

Solve each system using Gauss-Jordan Elimination

1)

$$\begin{aligned}x - y + z &= 2 \\2x + y - z &= 1 \\x + y + z &= 6\end{aligned}$$

2)

$$\begin{aligned}2x + y + z &= -2 \\x - 2y - z &= 2 \\3x - 6y - 3z &= 4\end{aligned}$$

Ex. 5: Find A^2 given $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 3 & 2 & 1 \end{bmatrix}$

Ex. 6: Given $A = \begin{bmatrix} 1 & 0 & 3 \\ -1 & 4 & -3 \\ 3 & 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 4 & 6 \\ 0 & 2 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & -1 & 6 \\ 0 & 3 & \pi \end{bmatrix}$, find (if possible)

a) AC

b) CA

c) AB

c) $Z = \begin{bmatrix} 1 & 2 \\ 5 & 9 \end{bmatrix}$

d) $M = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}$

e) $C = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ -2 & 3 & -2 \end{bmatrix}$

Ex. 2: Find the determinant of $\begin{bmatrix} 5 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$.

Ex. 3: Find the determinant of

a) $\begin{bmatrix} 0 & 3 & 8 \\ 3 & -2 & 2 \\ -7 & 5 & 0 \end{bmatrix}$

b) $\begin{bmatrix} 0 & 2 \\ 3 & 8 \end{bmatrix}$

c) $\begin{bmatrix} 2 \sin \theta & -2 \cos \theta \\ 2 \cos \theta & 2 \sin \theta \end{bmatrix}$